

Biology Study Guide: Photosynthesis

Photosynthesis is the process used by plants, algae, and certain bacteria to harness energy from sunlight and turn it into chemical energy.

The Chloroplast

Photosynthesis takes place inside organelles called **chloroplasts**. Chloroplasts contain a green pigment called **chlorophyll**, which absorbs light energy.

Two Stages of Photosynthesis

Photosynthesis occurs in two distinct stages:

- **Light-dependent Reactions:** These reactions occur in the thylakoid membranes and require direct light energy. Water molecules are split to release oxygen gas, generating ATP and NADPH.
- **Light-independent Reactions (Calvin Cycle):** These reactions occur in the stroma and do not require light. They use ATP and NADPH from the light-dependent reactions to fix carbon dioxide into glucose.

Chemical Equation

The balanced chemical equation for photosynthesis is:

